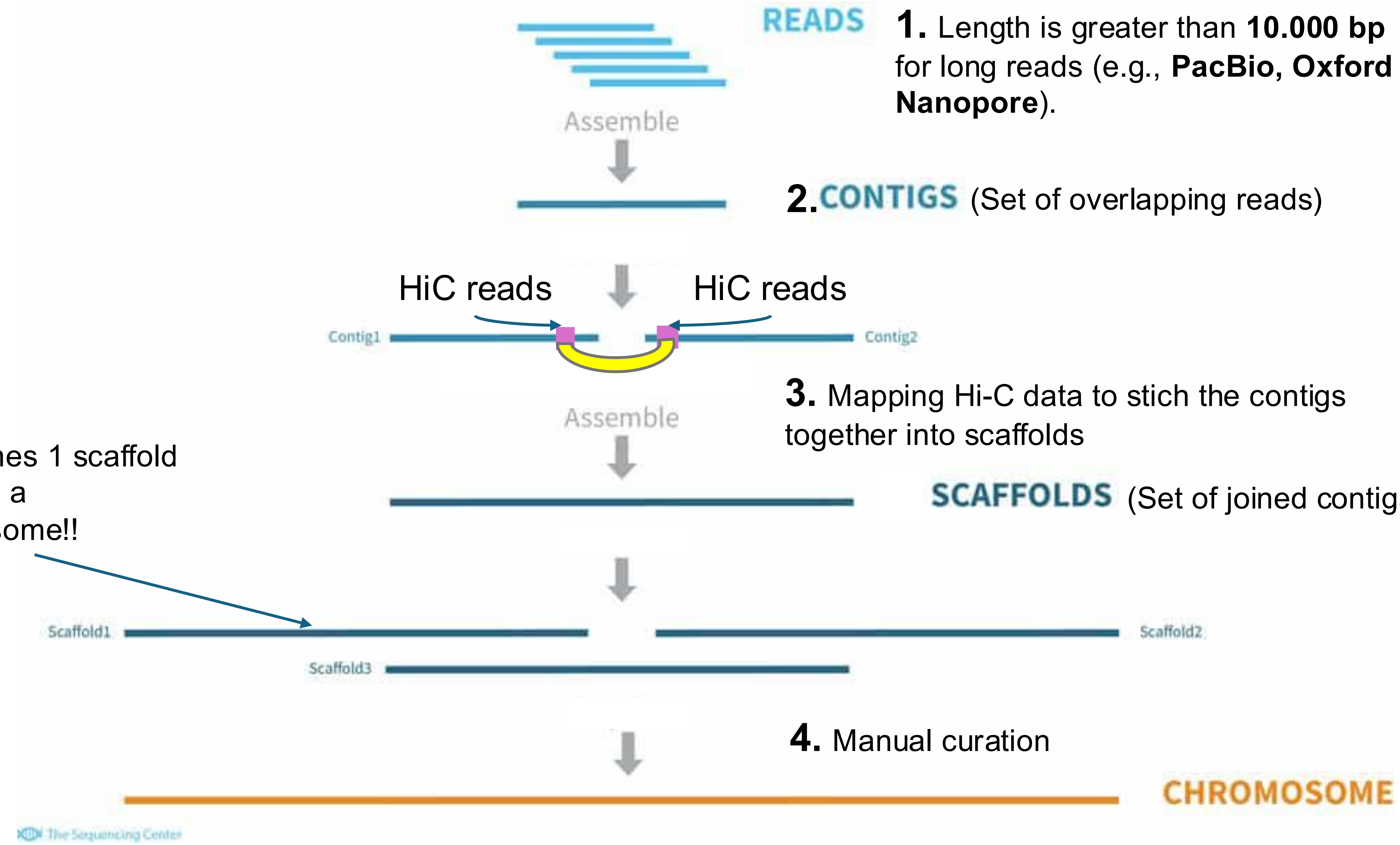
Genome assembly
Biodiversity genomics course
Mendoza-Argentina
2025

General steps genome assembly



1. Length is greater than **10.000 bp** for long reads (e.g., **PacBio, Oxford Nanopore**).

2. CONTIGS (Set of overlapping reads)

HiC reads

HiC reads

3. Mapping Hi-C data to stich the contigs together into scaffolds

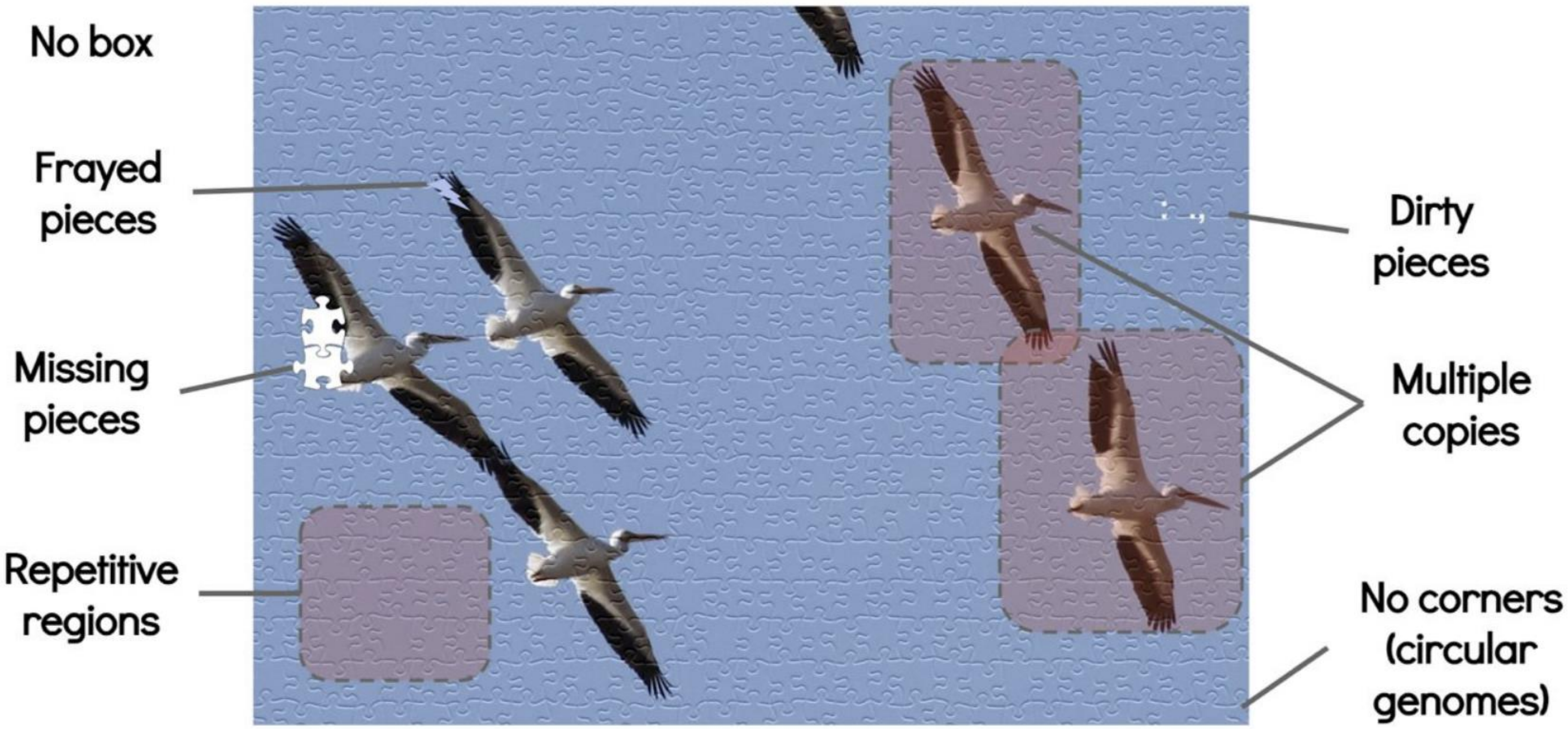
SCAFFOLDS (Set of joined contigs)

Sometimes 1 scaffold could be a chromosome!!

4. Manual curation

Why is important to use long reads?

What makes a jigsaw puzzle hard?

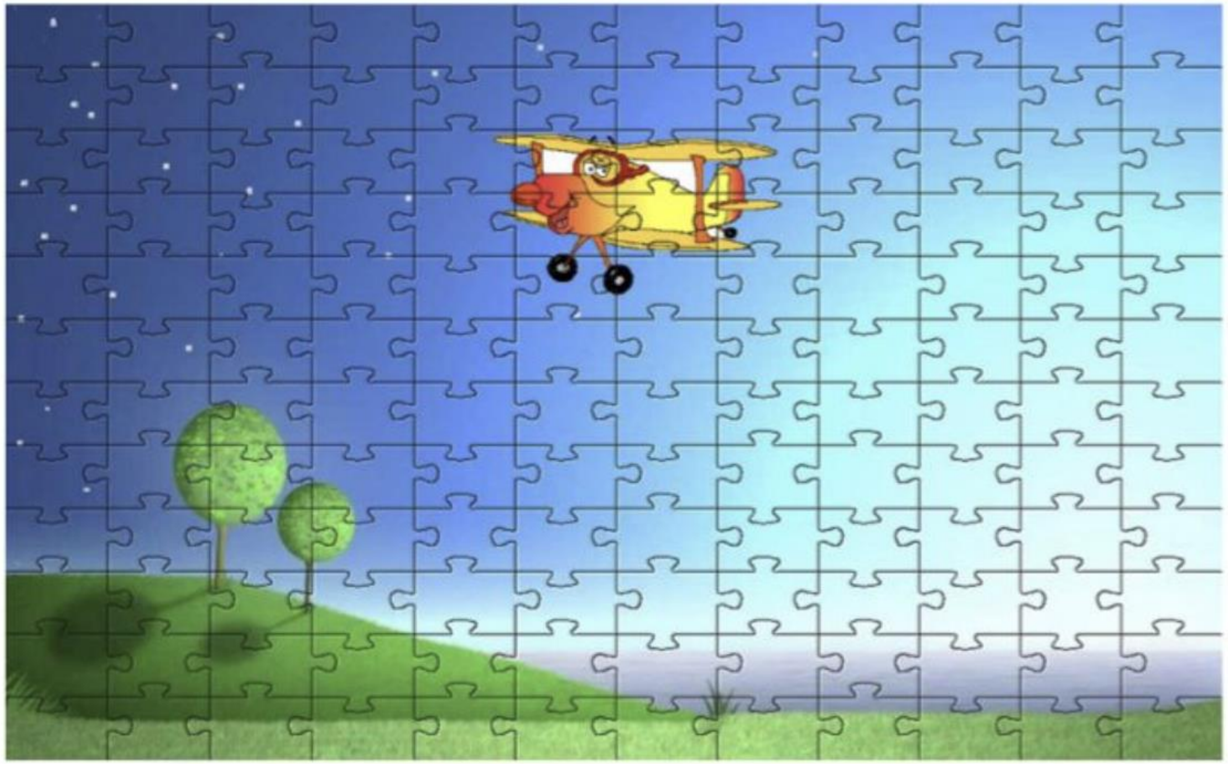


- What helps? Larger pieces (read length); fewer dirty or frayed pieces (errors in reads). fewer repeats and copies…

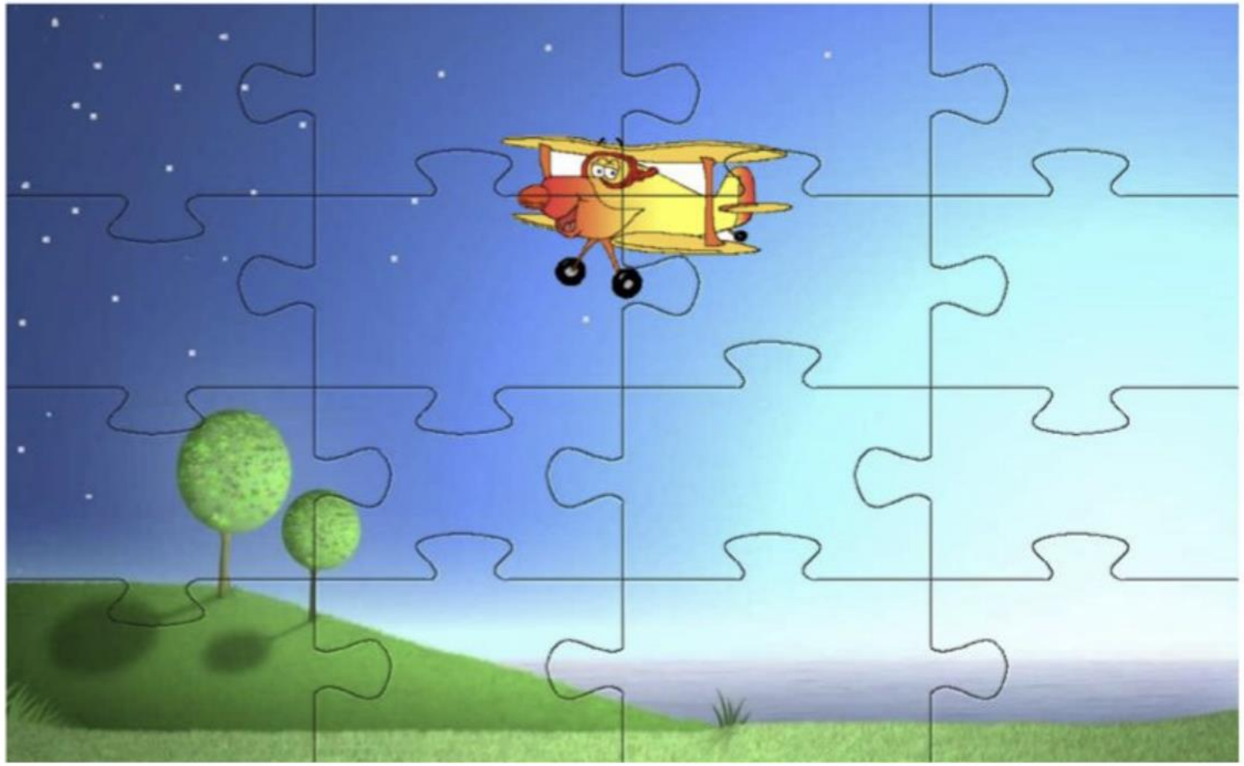
Why is important to use long reads?

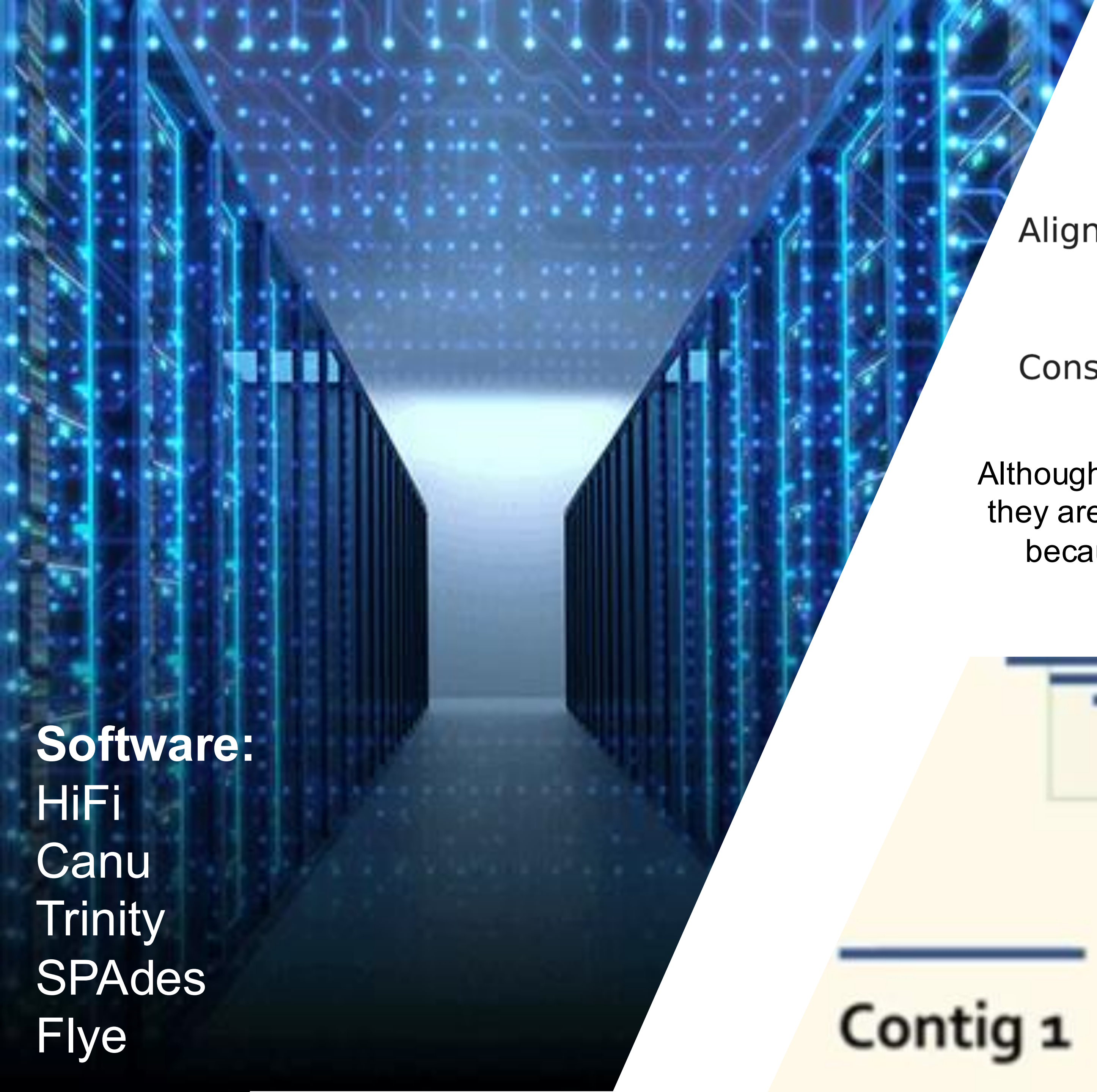
PacBio, Oxford Nanopore

Genome assembly with short reads



Genome assembly with long reads





Software:
HiFi
Canu
Trinity
SPAdes
Flye

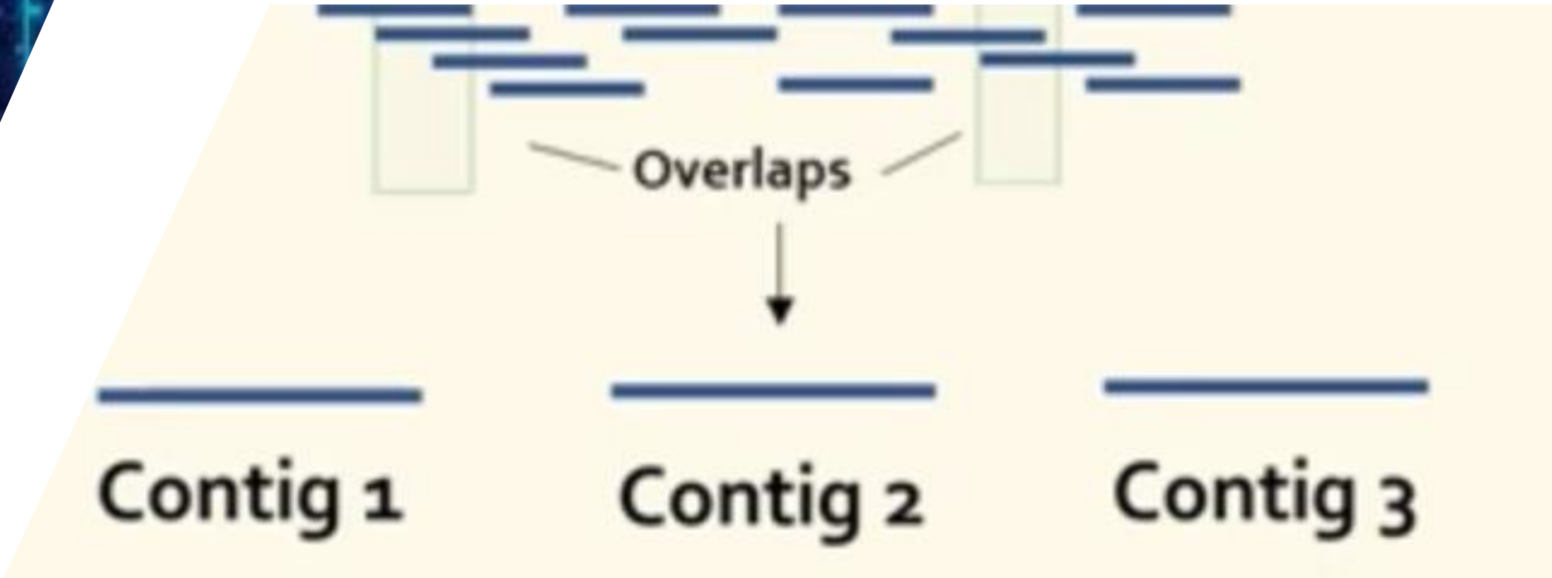
Step 2. Assembling the reads

Aligned reads

```
TGAAGTCCTACAGTCATAGTC
   AAGTCCTACAGTCATAGTCGA
      GTCCTACAGTCATAGTCGATA
         CCTACAGTCATAGTCGATATT
            TACAGTCATAGTCGATATTT
```

Consensus contig TGAAGTCCTACAGTCATAGTCGATATTT

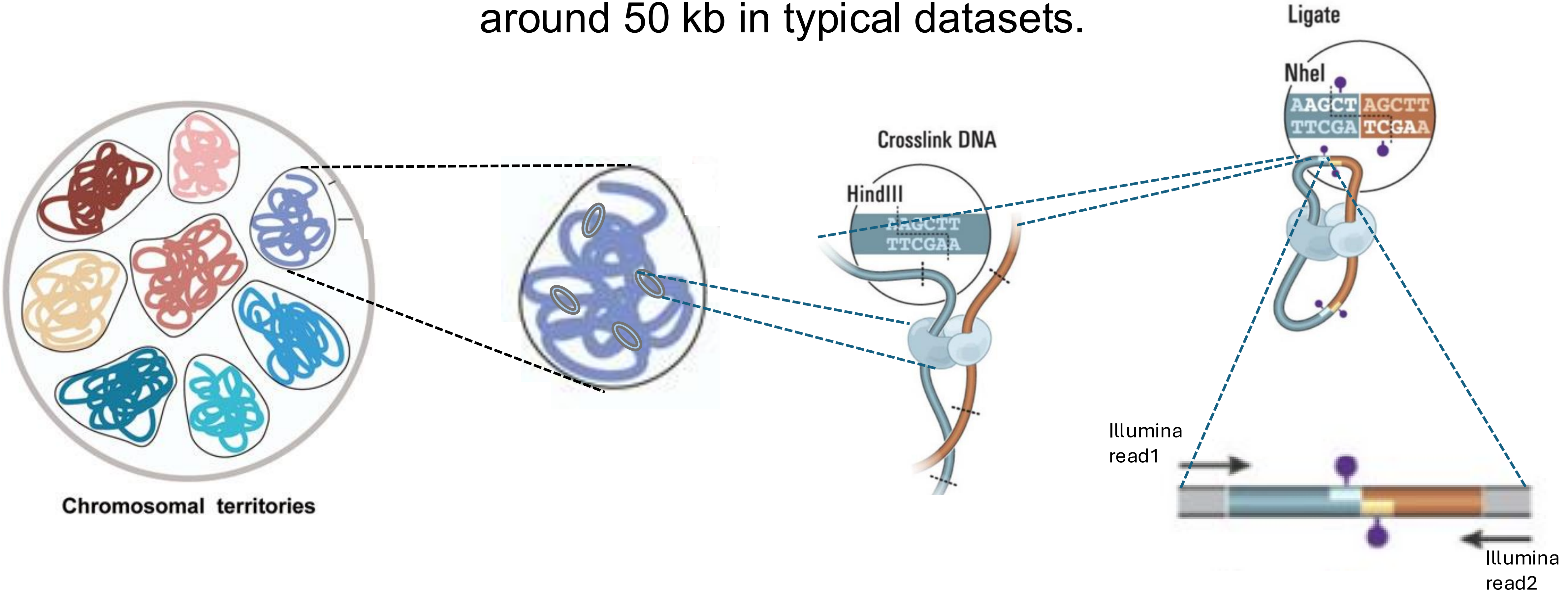
Although the contigs are longer than reads and do not have gaps, they are typically not large enough to cover entire chromosomes because some parts of chromosomes are very repetitive or otherwise hard to sequence trough.



Step 3. Scaffolding (Assembling the contigs)

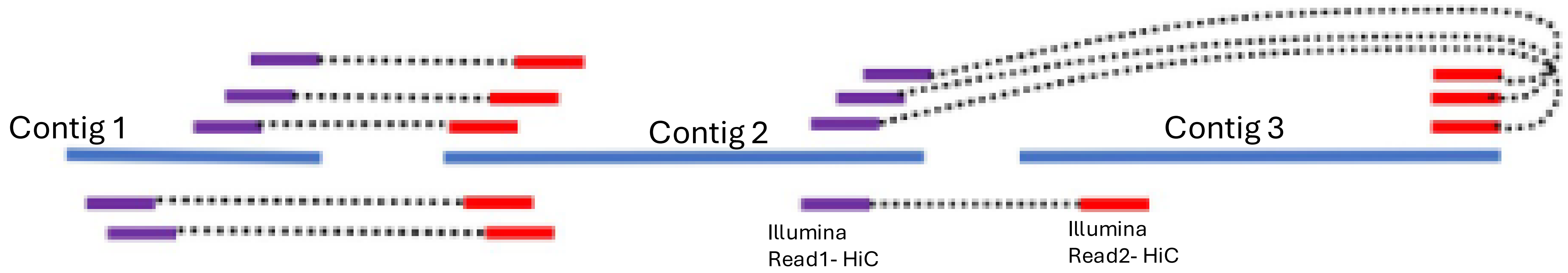
Hi-C data

Aim: Join two nearby parts of a chromosome together and then sequence across the join to get reads from regions of the same chromosome, often tens to hundreds of kilobases apart—for example, around 50 kb in typical datasets.



Step 3. Scaffolding (Assembling the contigs)

1. Mapping HiC data to Contigs:

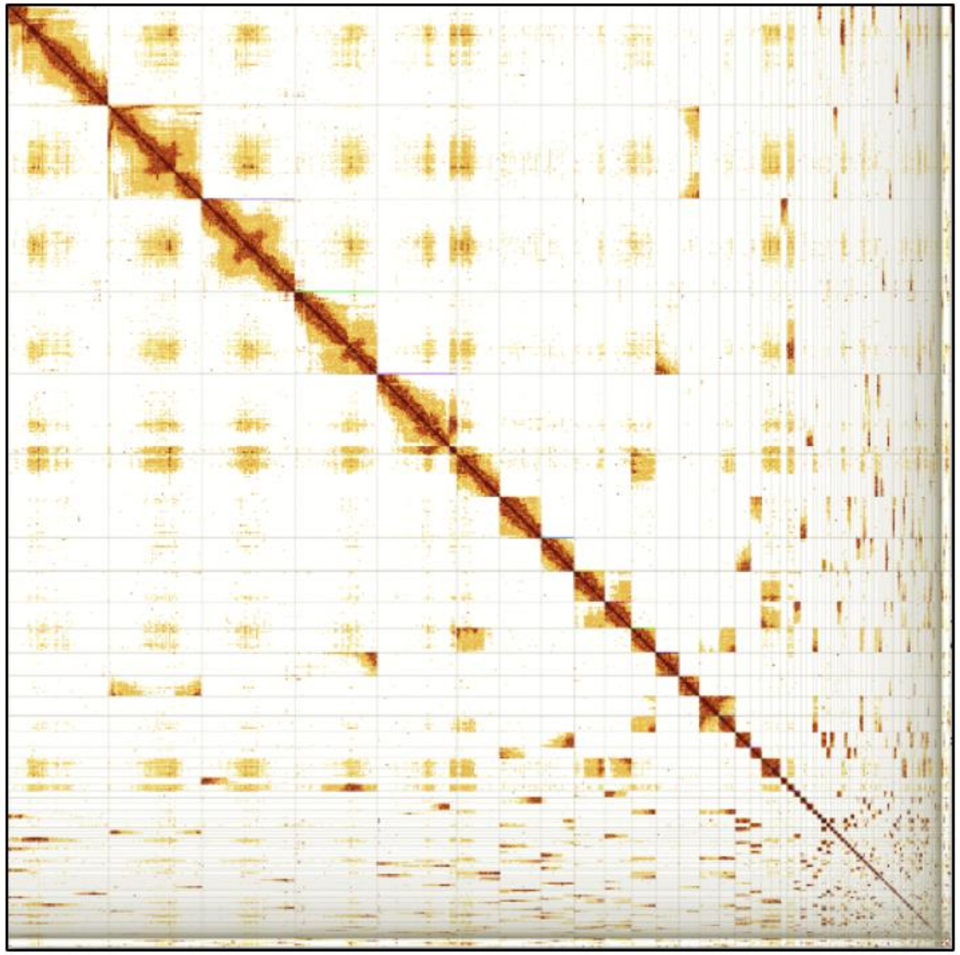


2. Stitching contigs together to form scaffolds:

Scaffold/Chromosome

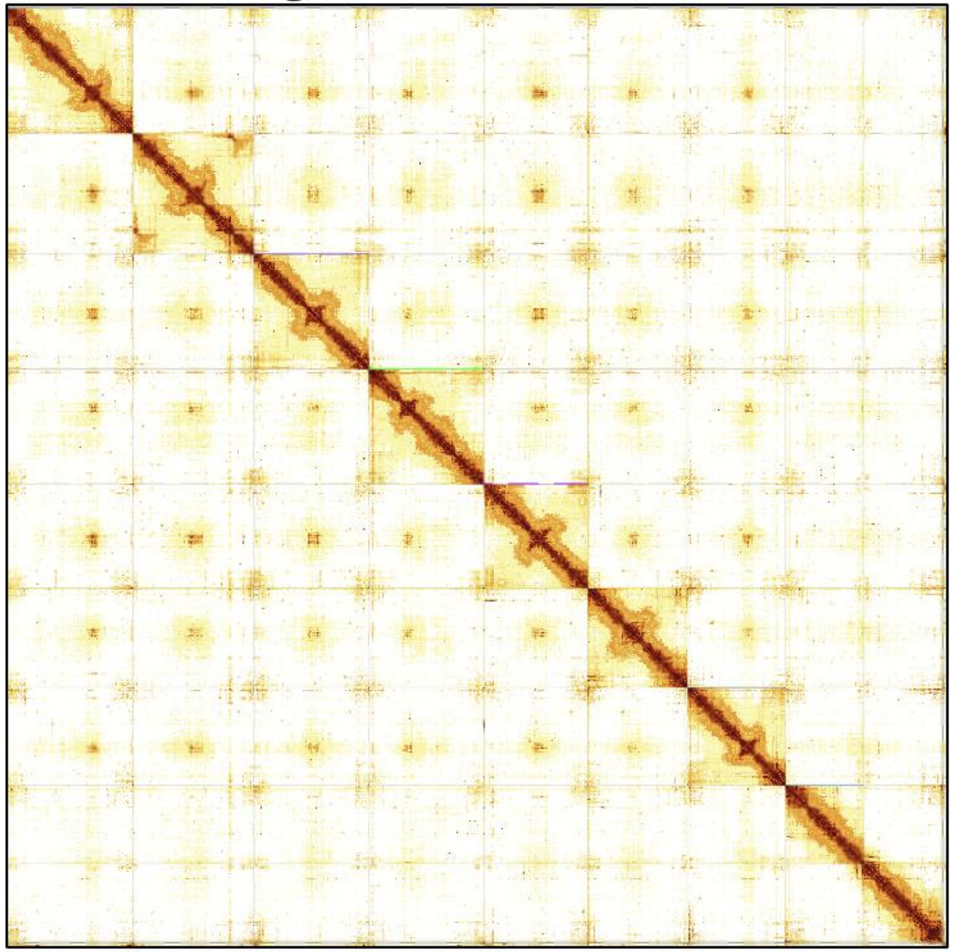
Step 4. Genome curation

n = 230
N50 = 33.1Mb



Pre curation assembly

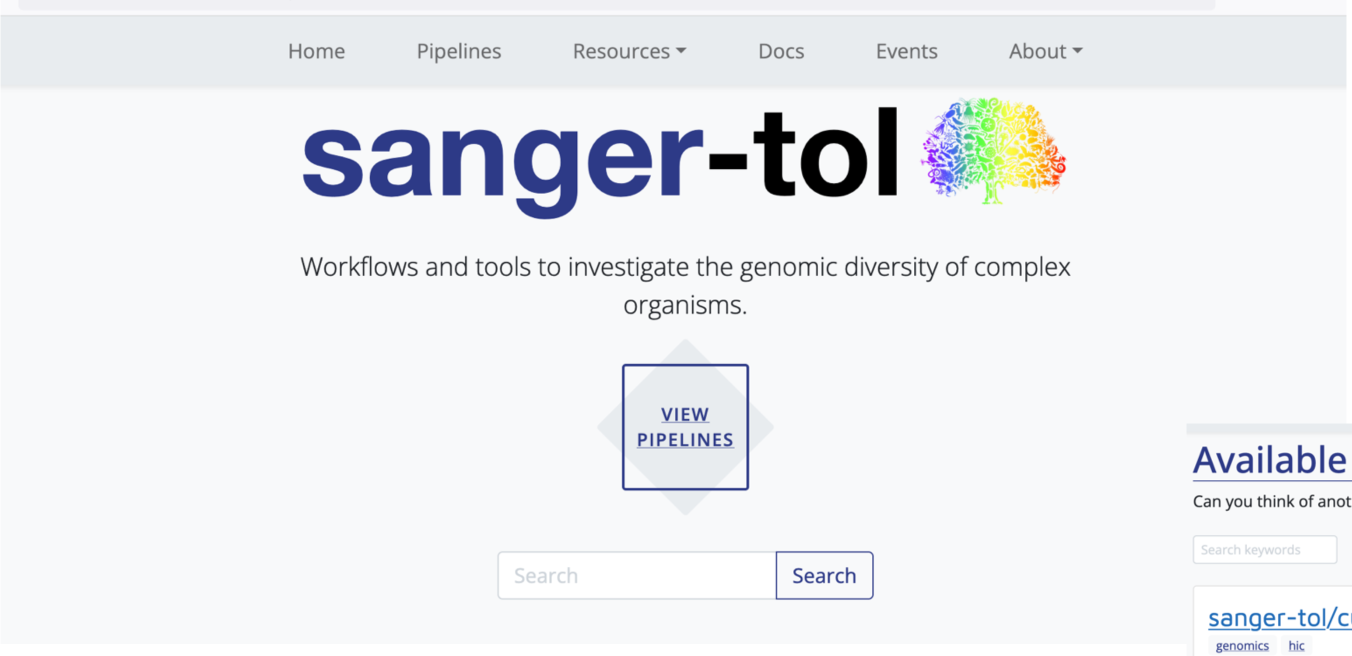
n = 62
N50 = 87.1Mb
99.85% of genome in 9 Chromosomes



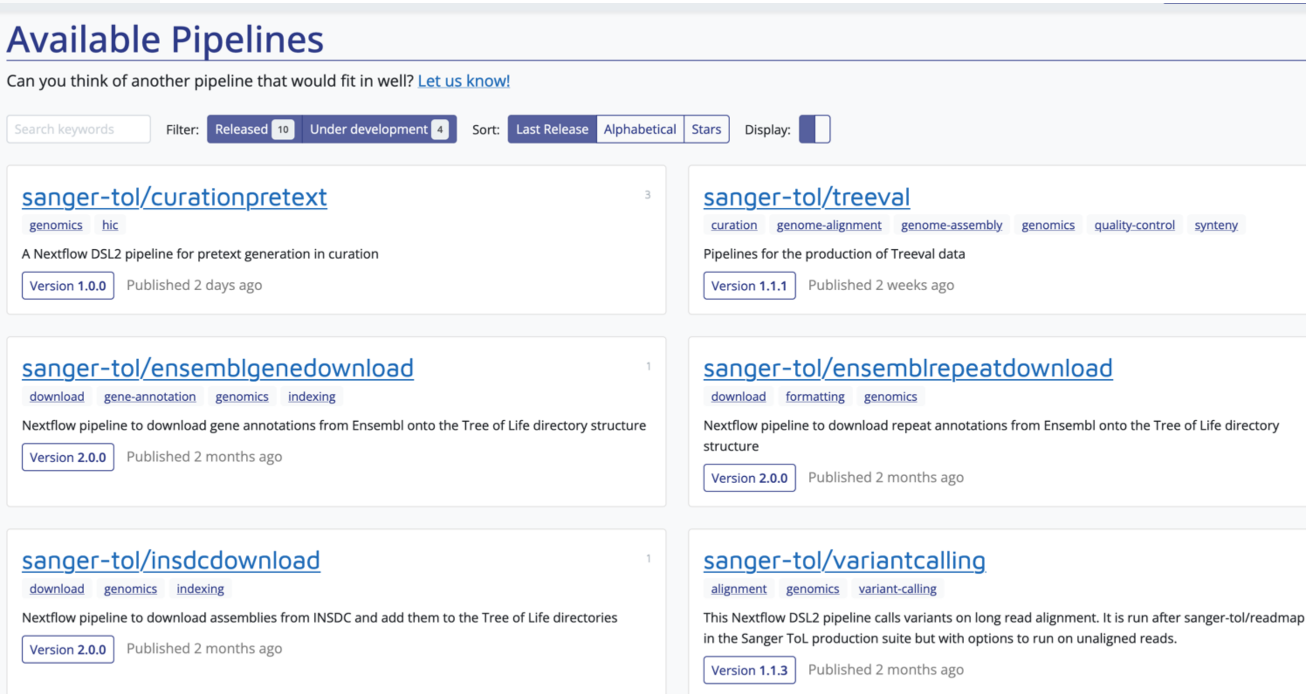
Post curation

Pipelines to generate a genome

https://pipelines.tol.sanger.ac.uk/



Sanger-Tol recipe is working across the Tree of life
4000 species



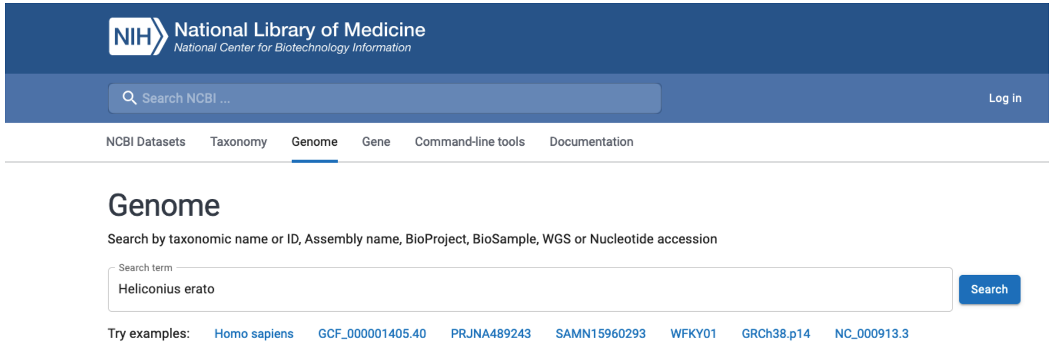
Chromosome level genomes

- 25x Pacbio HiFi
- 100x Hi-C (Arima/Qiagen)

Where can we find genomes?

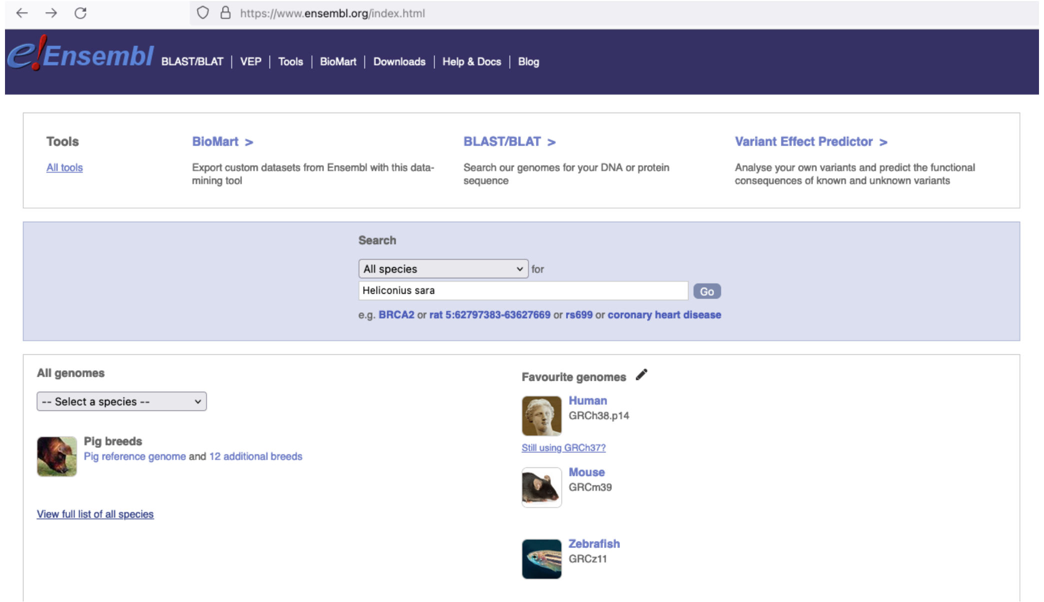
- NCBI (National Center for Biotechnology Information):

https://www.ncbi.nlm.nih.gov/datasets/genome/



- ensembl

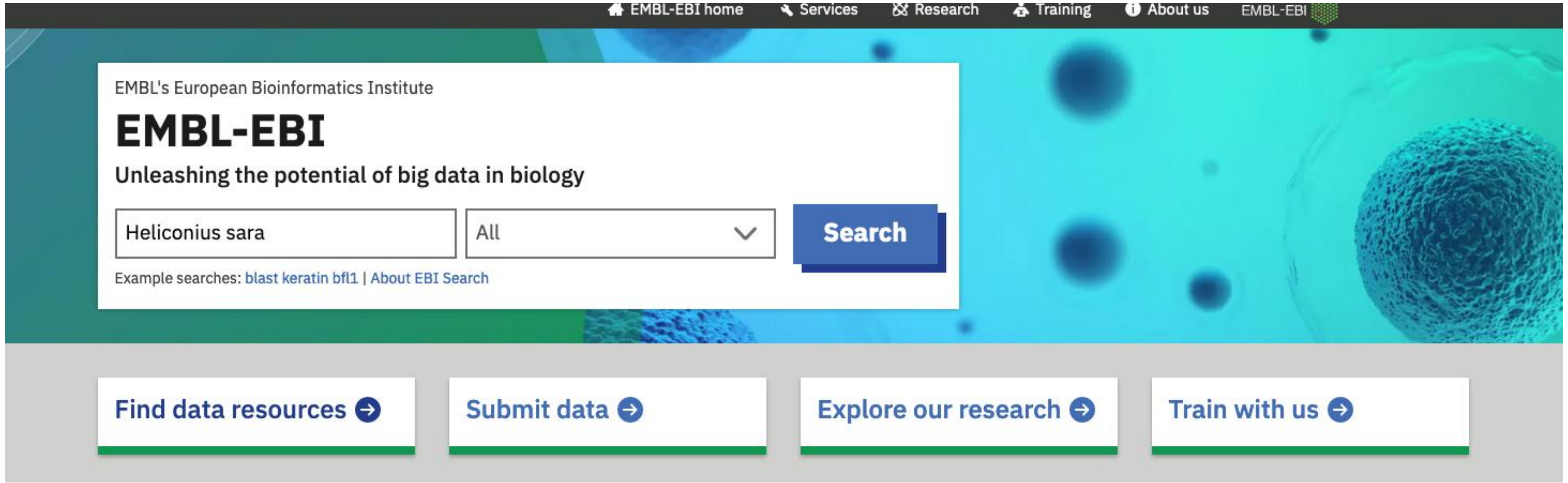
https://www.ensembl.org/index.html



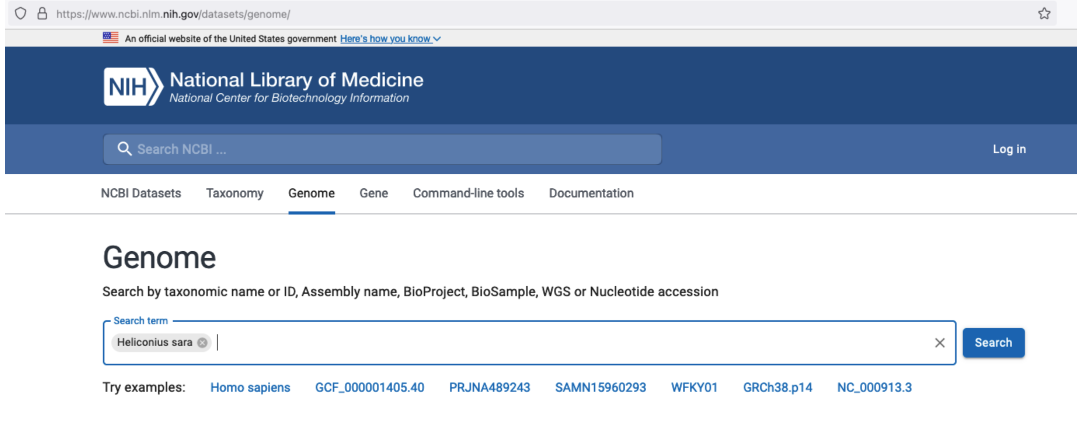
Where can we find genomes?

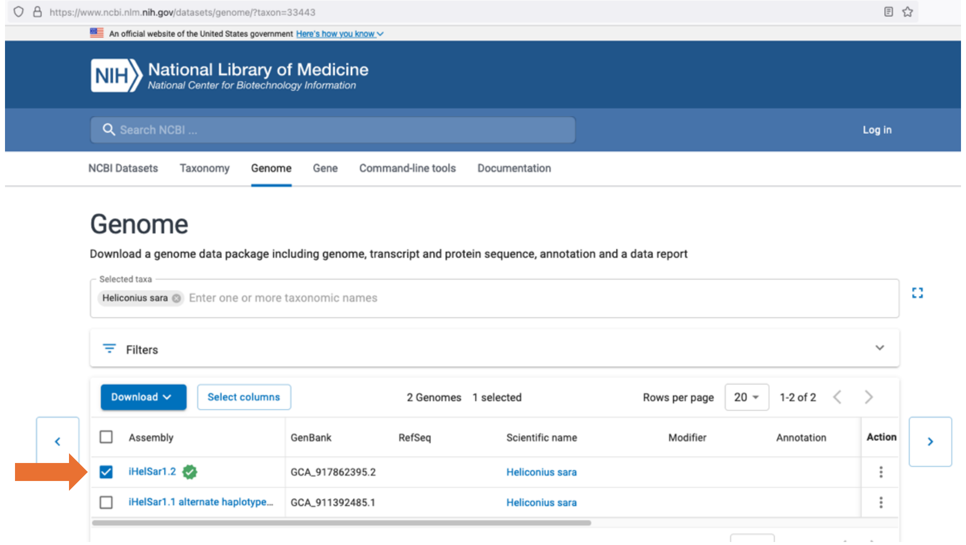
➢ EMBL-EBI (European Molecular Biology Laboratory - European Bioinformatics Institute):

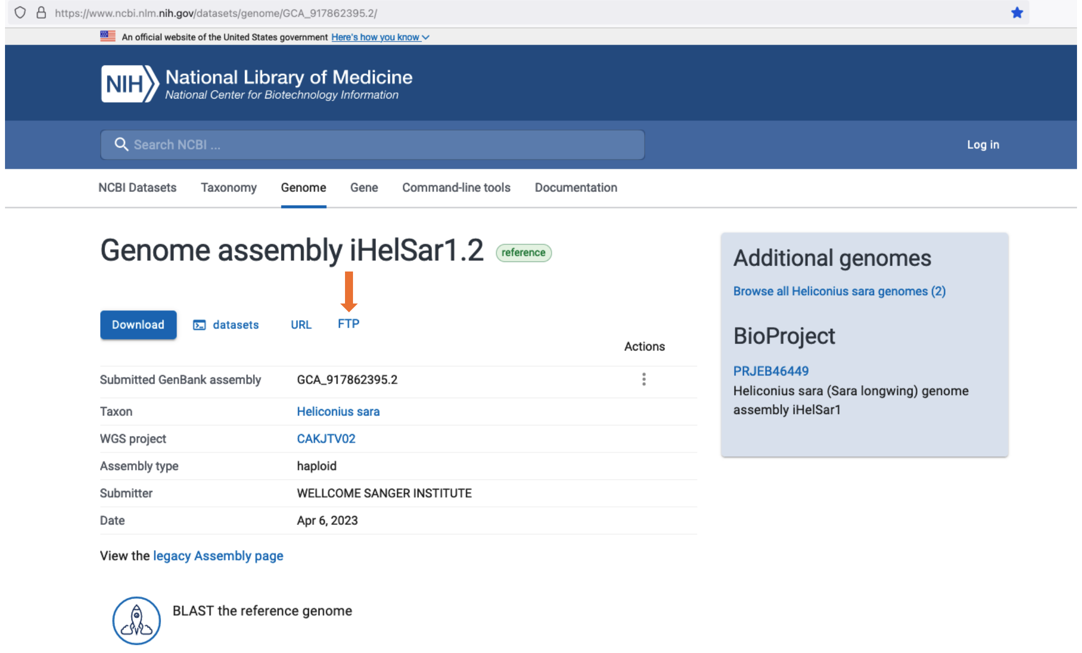
https://www.ebi.ac.uk/

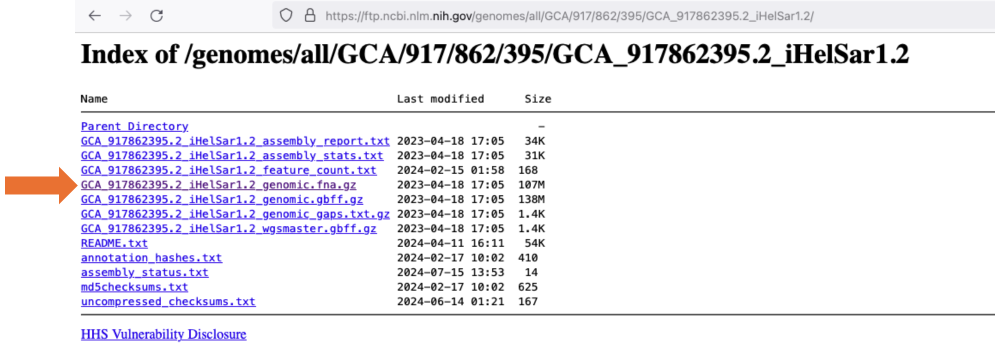


Example










```
wget
https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/917/862/395/GCA_917862395.2_iHelSar1.2/GCA_917862395.2_iHelSar1.2_genomic.fna.gz
```

Reference genomes are stored in a fasta format

It is a text-based format for representing nucleotide sequences (DNA or RNA)

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCTCTTTTCTTATCATTGACATTTAAACTCTGGGGCAGGTCCTCGCGTAGAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCCCGGGCTCCGGCCCCGGCCCGGCTCGGGGCCCGCGGGGCCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGTGCTCTTCTGCAAAAGTAGCAAAATGTTCCACTCCTAAGAGTGGACTTCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTCACGACATCCACGCTTGGGAAAG
TCCGTACCCGCGCCTGGAGCGCTTAAAGACACCCTGCCGCGGTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

Description Line

Sequence Lines

Description Line: Begins with a > character followed by an identifier and optional description.

Sequence Lines: The actual sequence data, typically represented over multiple lines for readability.

Assembly statistics

	GenBank
Genome size	562.2 Mb
Total ungapped length	562.2 Mb
Number of contigs	203,084
Contig N50	9.9 kb
Contig L50	14,088
GC percent	33.5
Genome coverage	1.9x
Assembly level	Contig

It is the median of the contig lengths.

It is the number of contigs (or scaffolds) that together contain at least 50% of the total assembly length.

Significance

N50: A higher N50 value indicates that the assembly contains longer contigs, suggesting better assembly quality and continuity.

L50: A lower L50 value indicates that fewer contigs are needed to cover 50% of the genome, suggesting higher assembly contiguity.

References

- Jay Ghurye and Mihai PopI. 2019. Modern technologies and algorithms for scaffolding assembled genomes. Plos computational biology
- Edward S. Rice and Richard E. Green. 2018. New Approaches for Genome Assembly and Scaffolding. Annual Review of Animal Biosciences.
- https://ucdavis-bioinformatics-training.github.io/2020-Genome_Assembly_Workshop/kmers/kmers
- https://www.youtube.com/watch?v=8w-CbyGV-pgv
- https://star-protocols.cell.com/protocols/1799
- https://training.galaxyproject.org/training-material/topics/sequence-analysis/tutorials/quality-control/tutorial.html#per-sequence-quality-scores